

TableMorph: Haptic Experience with Movable Tables and Redirection

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Figure 1: Left: Concave and convex self-propelled tables. Center: TableMorph can reproduce arbitrarily shaped tables by means of Encounter-type haptics and Redirection. Right: A user touching the tables.

ABSTRACT

We presented “TableMorph,” a novel method that combines redirection techniques and encountered-type haptic displays. In this method, concave and convex-shaped tables with mobile robots move through the real environment according to the relative positions of the user and the virtual tables in the virtual environment. In the process, by using redirection techniques that change the position and orientation of the user’s virtual hand and the shape of the virtual tables visually presented through the head-mounted display from the actual one, the system can make them feel, both visually and haptically, that the table has a shape very different from the real one. In the demonstration at SIGGRAPH ASIA 2023 E-tech, participants can try virtual mazes as an application of the proposed method. As the users move around in the virtual maze separated by tables, the arrangement of the tables in the real environment changes. This allows for the presentation of various types of mazes that are four times larger than the physical space.

CCS CONCEPTS

• **Human-centered computing** → **Virtual reality**; *Haptic devices*.

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KEYWORDS

Encountered-type Haptics Display, Redirected Hand, Haptic Retargeting, Hand Redirection, Redirected Walking, Virtual Reality

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1 INTRODUCTION

In virtual reality (VR), numerous research studies have tackled the challenge of reproducing haptic feedback for various virtual objects. One prominent approach is the encountered-type haptic (ETH) display. This approach aims to provide haptic feedback for virtual objects visually presented in the virtual environment by utilizing robotic systems to pre-position physical objects that users can physically interact with. This creates the illusion that the objects have been there all along. Notable examples of this technique include ZoomWalls [Yixian et al. 2020] and RoomShift [Suzuki et al. 2020]. However, ETH has several challenges, including the limited variety of haptic feedback that can be reproduced in the virtual environment compared to the haptic feedback provided by physical objects. Another challenge is that ETH cannot place haptic objects in virtual spaces larger than physical ones.

Meanwhile, recently another approach has been proposed to present haptics in VR called redirected hand (RDH, also known as redirected touching, haptic retargeting, or hand redirection.) RDH enables the user to interact with objects in different positions, assume various postures, or even exhibit different shapes than their

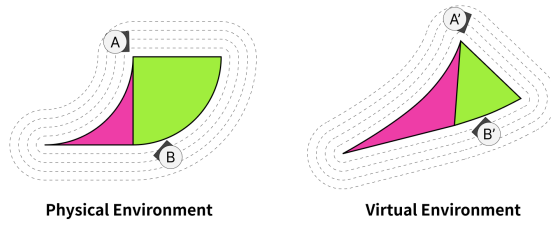


Figure 2: Combination of convex and concave tables. User A and B are redirected to A' and B'.

actual state. Magic Table [Matsumoto et al. 2017], which utilizes RDH, aims to address the limitations by altering the visual stimuli of a single object within the VR environment, enabling users to interact with various shapes. We have used the body warping method, one of the fundamental RDH methods [Zenner and Krüger 2019].

We have developed a novel system called TableMorph that combines ETH and RDH. In TableMorph, concave and convex-shaped tables (Figure 1 left) dynamically adjust their positions to align with the user's virtual experience. Additionally, by redirecting the user's hand and objects, TableMorph enables haptic feedback of tables with more flexible shapes (Figure 1 center and right). TableMorph can also place objects over a wider area by redirecting the relationship between the user and the object's position.

2 SYSTEM DESIGN

We created a set of tables with concave and convex shaped and straight edges equipped with movable robots. We also developed a program that applies different position and rotation redirection to the edges and corners of the table.

A set of concave and convex tables was designed by dividing a square table into two parts by a rectangle with one side as a radius as illustrated in the Figure 1 left. The length of each side of the table is 850 mm, and the height of the tables is 800 mm. Each table is equipped with a TURTLEBOT3 Burger, each of which can move forward and backward at 0.22 m/s and rotate on its own axis. The table position was determined by a vive tracker attached to the table.

A virtual world was created by Unity 2021.3.11f1. To present the virtual world for users, we used an HTC VIVE PRO as a head-mounted display. HP VR Backpack G2 did the VR rendering and Turtle bot control.

Using redirection techniques, the convex and concave tables were designed to allow for changes in edge lengths and internal angles in the virtual environments. We determined the virtual positions and orientations of the HMD and the hands for each table in correspondence to the relationship between the physical table and the physical HMD and trackers equipped on the hands in the physical space.

If two or more tables are present in the scene, it is necessary to integrate the calculated virtual positions and orientations of the HMD and hands from each table in order to determine the final virtual positions of the HMD and hands. In such cases, the contribution of each table can be calculated by considering parameters such as the distance between the HMD, trackers, and the physical tables. From this calculation, the weighted average position and

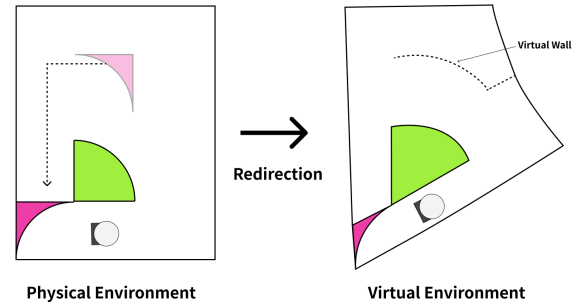


Figure 3: The physical and virtual maze maps. The user goes through the maze. The positions of the tables change as the user moves. The user is redirected differently from in the initial state.

orientation of the virtual HMD and hands are determined. If two or more tables are closely connected and share an edge or corner, they can be considered one, and redirection can be applied like Figure 2.

3 DEMONSTRATION AT SIGGRAPH ASIA

In the demonstration, participants attempt to escape from a virtual maze in a virtual environment approximately four times the size of the physical space. A participant wears an HMD and walks around in a physical space of 3 m × 4 m surrounded by walls on all sides, where convex and concave tables are arranged. The shape of the virtual maze dynamically changes depending on the participant's position using redirection techniques (Figure 3). The tables are utilized as ETH; when combined with redirection techniques, they become objects with various roles, from obstacles that block the path to objects that provide clues to escape from the maze. Interacting with the obstacles and cue objects provides an immersive and exciting experience.

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